

Spring Boot 3 JPA & Backend Summary

1. JPA & Hibernate Basics

Spring Boot 3 integrates with Hibernate for JPA implementation. Entities are annotated with `@Entity` and mapped to database tables. Repositories extend `JpaRepository` or `CrudRepository` for CRUD operations. Named queries, JPQL, Criteria API, and Specifications provide querying flexibility.

2. DTOs & Projections

To avoid N+1 and reduce payload, we use DTO projections (constructor-based, interface-based, or `@EntityGraph`). DTOs map selected fields, improving performance. `@SqlResultSetMapping` and `@NamedNativeQuery` support native queries returning DTOs.

3. Pagination & Sorting

Spring Data JPA provides `Pageable` and `Sort` abstractions. Pagination ensures efficient data fetching by limiting results. Sorting orders results dynamically. Example: `Page.findAll(Pageable pageable)`.

4. REST API Flow

Flow: Client → Controller → Service → Repository → Database. Controllers handle requests, delegate to services for business logic, and repositories handle persistence. DTOs ensure separation between entity and API response.

5. Interceptor Flow

Interceptors (`HandlerInterceptor` or `OncePerRequestFilter`) allow cross-cutting concerns: logging, authentication, metrics. Flow: Request → `DispatcherServlet` → Interceptor (`preHandle`) → Controller → Service → Repository → Response.

6. Kafka Integration

Spring Boot integrates with `KafkaTemplate` for producers and `@KafkaListener` for consumers. Flow: Producer Service sends message → Kafka Broker → Consumer Service processes event. Useful for event-driven architecture.

7. HATEOAS

HATEOAS enriches REST responses with navigable links. Spring HATEOAS provides `EntityModel`, `CollectionModel`, and `PagedModel` for hypermedia support. Combined with pagination, it improves discoverability of APIs.

8. Transaction Management

@Transactional ensures ACID properties. Supports propagation levels (REQUIRED, REQUIRES_NEW) and rollback policies. Transaction boundaries are usually at service layer. Programmatic TransactionTemplate is also available.

9. Specifications & Criteria API

Specifications (org.springframework.data.jpa.domain.Specification) allow dynamic query building with Predicates. Example: findAll(Specification spec). Useful for advanced filtering and search APIs.